BITS Pilani

Pilani|Dubai|Goa|Hyderabad

Introduction to Data Science

Data

- *The slides presented here are obtained from the authors of the books and from various other contributors. I hereby acknowledge all the contributors for their material and inputs.*
- *We have added and modified slides to suit the requirements of the course.*

Types of Attributes

Data

Data is a collection of data objects and their attributes.

An attribute is a property or characteristic of an object.

Examples: eye color of a person, temperature

Attribute is also known as variable, field, characteristic, or feature.

A collection of attributes describe an object.

Object is also known as record, point, case, sample, entity, or instance.

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

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Attribute/Feature

An attribute is a property or characteristic of an object.

- eye color of a person, temperature

Attribute is also known as variable, field, characteristic, or feature.

The values used to represent an attribute may have properties that are not properties of the attribute itself.

- Average age of an employee may have a meaning , whereas it makes no sense to talk about the average employee ID.

Attributes

Objects

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1	Yes	Single	125K	No
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Properties of Attributes

Specify the type of an attribute by identifying the properties that correspond to underlying properties of the attribute.

Properties include

- Distinctiveness =, /=
- Order <, >, ≥, ≤
- Addition +, −
- Multiplication ∗, /

Based on these properties, we define four types of attributes: nominal, ordinal, interval, and ratio.

Each attribute type possesses all of the properties and operations of the attribute types above it.

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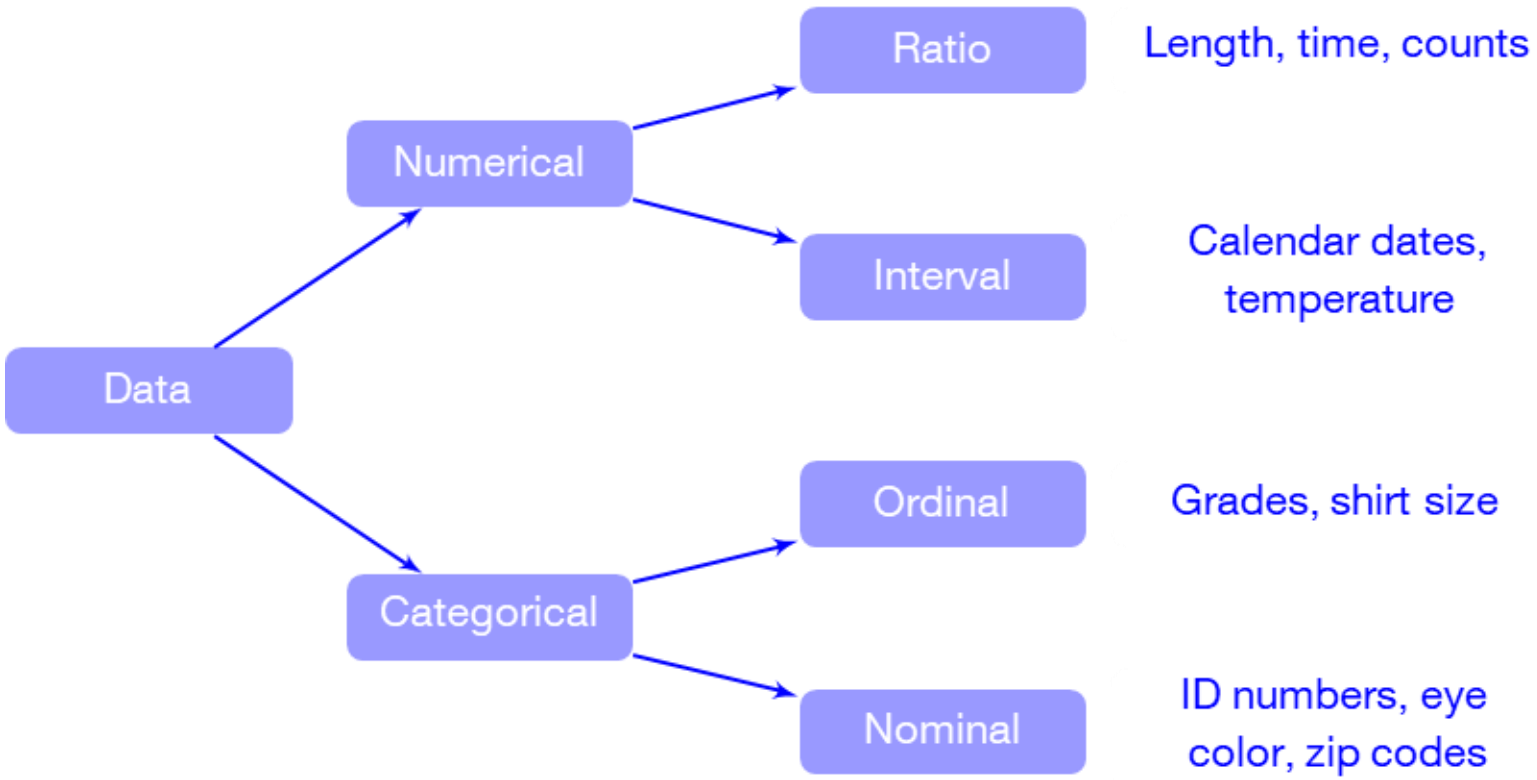
Types of Attributes



Types of Attributes

Levels of Measurement

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

Types of Attributes

Attribute Type		Description	Examples	Operations
Categorical (Qualitative)	Nominal	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another. $(=, \neq)$	zip codes, employee ID numbers, eye color, gender	mode, entropy, contingency correlation, χ^2 test
	Ordinal	The values of an ordinal attribute provide enough information to order objects. $(<, >)$	hardness of minerals, $\{good, better, best\}$, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Numeric (Quantitative)	Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. $(+, -)$	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
	Ratio	For ratio variables, both differences and ratios are meaningful. $(*, /)$	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	geometric mean, harmonic mean, percent variation

Discrete and Continuous Attributes

Discrete Attribute

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables.
- Note: binary attributes are a special case of discrete attributes

Continuous Attribute

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can only be measured and represented using a finite number of digits.
- Continuous attributes are typically represented as floating-point variables.

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Asymmetric Attributes

Asymmetric Attribute

- Only presence (a non-zero attribute value) is regarded as important
- Examples:
  - Words present in documents
  - Items present in customer transactions
  - Subjects taken by a student

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Types of Attributes Example

Identify the types of attributes in the given data.

ID	Age	Gender	Course ID	Percentage	Grade
19001	24	Female	CS 104	74	Good
19002	23	Male	CS 102	75	Good
19003	25	Female	CS 103	67	Fair
19004	24	Female	CS 104	79	Good
19005	23	Male	CS 102	75	Good
19006	24	Female	CS 103	87	Excellent
19007	26	Male	CS 105	70	Good

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19006	24	Female	CS 103	87	Excellent
19007	26	Male	CS 105	70	Good
Nominal	Ratio	Nominal	Nominal	Ratio	Ordinal

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Types of Attributes Example

Identify whether the attribute is discrete and continuous in the given data.

ID	Age	Gender	Course ID	Percentage	Grade
19001	24	Female	CS 104	74	Good
19002	23	Male	CS 102	75	Good
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Types of Attributes Example

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19006	24	Female	CS 103	87	Excellent
19007	26	Male	CS 105	70	Good
Discrete	Continuous	Discrete	Discrete	Continuous	Discrete

Types of Data-sets

Types of Data-sets

Structured data

- Data containing a defined data type, format and structure.
- Example: transaction data, online analytical processing , OLAP data cubes, traditional RDBMS, CSV file and spreadsheets.

Semi structured data

- Textual data file with discernible pattern that enables parsing
- Example: XML data file, HTML of a web page

Unstructured data

- Data that has no inherent structure.
- Example: text document, PDF, images and video, email

We may also have

Quasi structured data

- Textual data with erratic data format that can be formatted with effort, tools and time
- Example: Web click-stream data

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Structured Data

id	name	age
1	Jim	28
2	Pam	26
3	Michael	42

id	subject	Teacher
1	Languages	John Jones
2	Track	Wally West
3	Swimming	Arthur Curry
4	Computers	Victor Stone

student_id	subject_id	grade
2	1	98
1	2	100
1	4	75
3	3	60
2	4	76
3	2	88

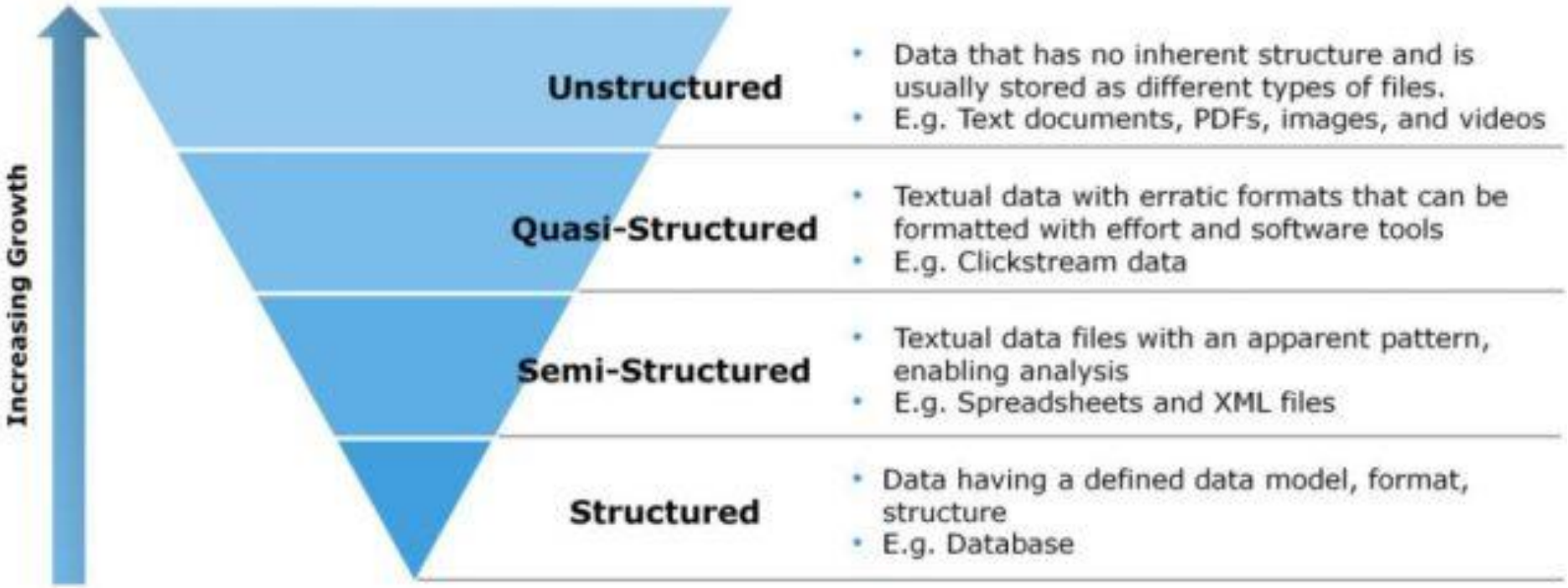
RDBMS Data

Semi-Structured Data

JSON Data

```
JSON Object → {                                  String Value
                                                 ↓
  "company": "mycompany",
  "companycontacts": { ← Object Inside Object
    "phone": "123-123-1234",
    "email": "myemail@domain.com"
  },
  "employees": [ ← JSON Array
    {
      "id": 101,
      "name": "John",
Array Inside Array → "contacts": [
        "email1@employee1.com",
        "email2@employee1.com"
      ]
    },
    {
      "id": 102, ← Number Value
      "name": "William",
      "contacts": null ← Null Value
    }
  ]
}
```

Types of Digital Data



Types of Digital Data, Pic Credits: EMC

Datasets

Public data

Data that has been collected and preprocessed for academic or research purposes and made public.

https://archive.ics.uci.edu/
https://www.kaggle.com/
https://search.open.canada.ca/opendata
Etc.

Private data

Data that is specific to an organization.

Privacy rules like IT Act 2000 and GDPR applies.

Record Data

Data that consists of a collection of records, each of which consists of a fixed set of attributes

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
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3	No	Single	70K	No
4	Yes	Married	120K	No
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7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
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10	No	Single	90K	Yes

Data Matrix

If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute

Such a data set can be represented by an *m* by *n* matrix, where there are *m* rows, one for each object, and *n* columns, one for each attribute

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document Data

Each document becomes a ‘term’ vector

- Each term is a component (attribute) of the vector
- The value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Transaction Data

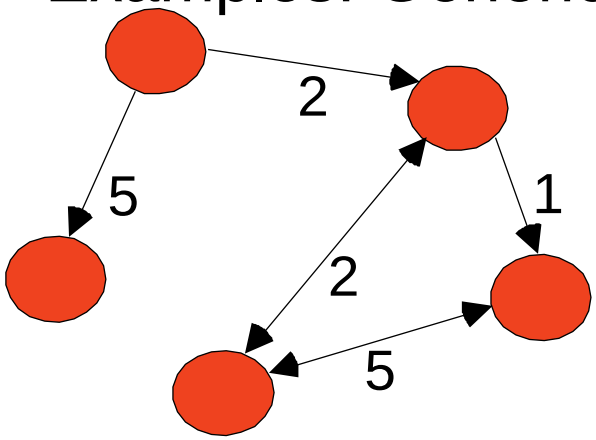
A special type of data, where

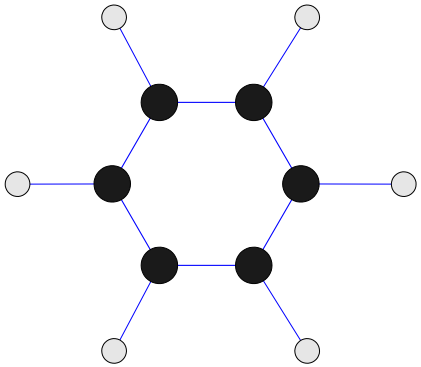
- Each transaction involves a set of items.
- For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.
- Can represent transaction data as record data

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Graph Data

Examples: Generic





Benzene Molecule: C6H6

Useful Links:

- Bibliography
- Other Useful Web sites
  - ACM SIGKDD
  - KDnuggets
  - The Data Mine

Knowledge Discovery and Data Mining Bibliography

(Gets updated frequently, so visit often!)

- Books
- General Data Mining

Book References in Data Mining and Knowledge Discovery

Usama Fayyad, Gregory Piatetsky-Shapiro, Padhraic Smyth, and Ramasamy uthurasamy, "Advances in Knowledge Discovery and Data Mining", AAAI Press/the MIT Press, 1996.

J. Ross Quinlan, "C4.5: Programs for Machine Learning", Morgan Kaufmann Publishers, 1993. Michael Berry and Gordon Linoff, "Data Mining Techniques (For Marketing, Sales, and Customer Support), John Wiley & Sons, 1997.

General Data Mining

Usama Fayyad, "Mining Databases: Towards Algorithms for Knowledge Discovery", Bulletin of the IEEE Computer Society Technical Committee on data Engineering, vol. 21, no. 1, March 1998.

Christopher Matheus, Philip Chan, and Gregory Piatetsky-Shapiro, "Systems for knowledge Discovery in databases", IEEE Transactions on Knowledge and Data Engineering, 5(6):903-913, December 1993.

Ordered Data

Sequences of transactions

Items/Events

(A B) (D) (C E)
(B D) (C) (E)
(C D) (B) (A E)

An element of the sequence

Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

Audio Data

Audio is considered as 1-D time series data Eg: Speech, Music

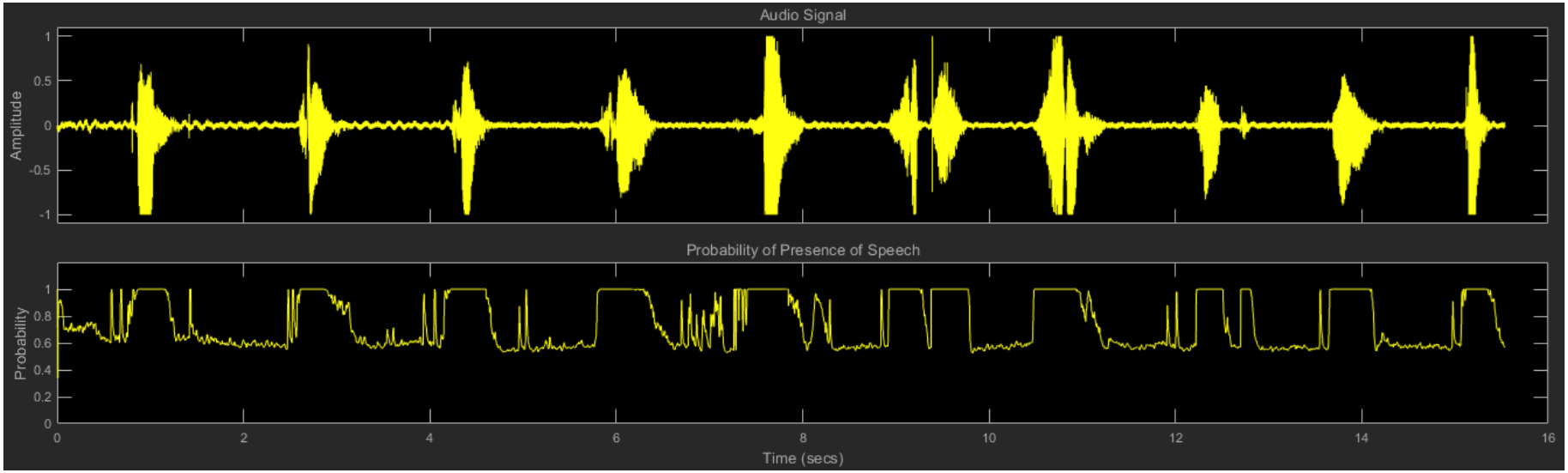


Image Data

Images are considered as 2-D data in Euclidean space

Digital Images are stored in a matrix or grid form where the intensity or colour information is stored in the (x,y) position.

Black and white image – intensity is represented as 0 and 1 respectively

Greyscale image – intensity is represented as an integer between 0 and 255. 0 represents black, grey is 125 and 255 is white.

Colour image – contains 3 bands or channels – Red, Green and Blue – each colour is represented as an integer between 0 and 255.

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Data Quality

Data Quality Issues

- Measures for data quality: A multidimensional view
  - Accuracy: correct or wrong, accurate or not
  - Completeness: not recorded, unavailable, …
  - Consistency: some modified but some not, dangling, …
  - Timeliness: timely update?
  - Believability: how trustable the data are correct?
  - Interpretability: how easily the data can be understood?

Data Quality Issues

- Missing data
  - Data that is not filled / available intentionally or otherwise.
  - Attributes of interest may not always be available, such as customer information for sales transaction data.
  - Some data were not considered important at the time of entry.
  - Relevant data may not be recorded due to a misunderstanding or because of equipment malfunctions.
- Duplicate data
- Orphaned data
- Text encoding errors
- Data that is biased

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Data Quality Issues

- Noise and outliers
  - Noise is a random error or variance in a measured data object.
  - Data objects with behaviors that are very different from expectation are called outliers or anomalies.
- Inaccurate data
  - Inaccurate data – data having incorrect attribute values
  - Caused by, faulty data collection instruments, human or computer errors occurring at data entry, users may purposely submit incorrect data values, errors in data transmission
- Inconsistent data
  - inconsistencies in naming conventions or data codes, or inconsistent formats for input fields (e.g., date).

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Example: Data quality Issues

Find the issues in the given data.

Name	Age	Date of Birth	Course ID	Percentage
Amy	24	01-Jan-1995	CS 104	74
Ben	23	Dec-01-1996	CS 102	75
Cathy	25	01-Nov-1994		67
Diana	24	Oct-01-1995	CS 104	79
Ben	23	Dec-01-1996	CS 102	75
Eden	24		CS 103	175
Fischer		01-01-1959	CS 105	70

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Example: Data quality Issues

Name	Age	Date of Birth	Course ID	Percentage
Amy	24	01-Jan-1995	CS 104	74
Ben	23	Dec-01-1996	CS 102	75
Cathy	25	01-Nov-1994		67
Diana	24	Oct-01-1995	CS 104	79
Ben	23	Dec-01-1996	CS 102	75
Eden	24		CS 103	175
Fischer		01-01-1959	CS 105	70

Missing data – age, date of birth, course ID

Inconsistent data – date of birth

Duplicate data – Ben is duplicated Data

Conformity – Percentage = 175

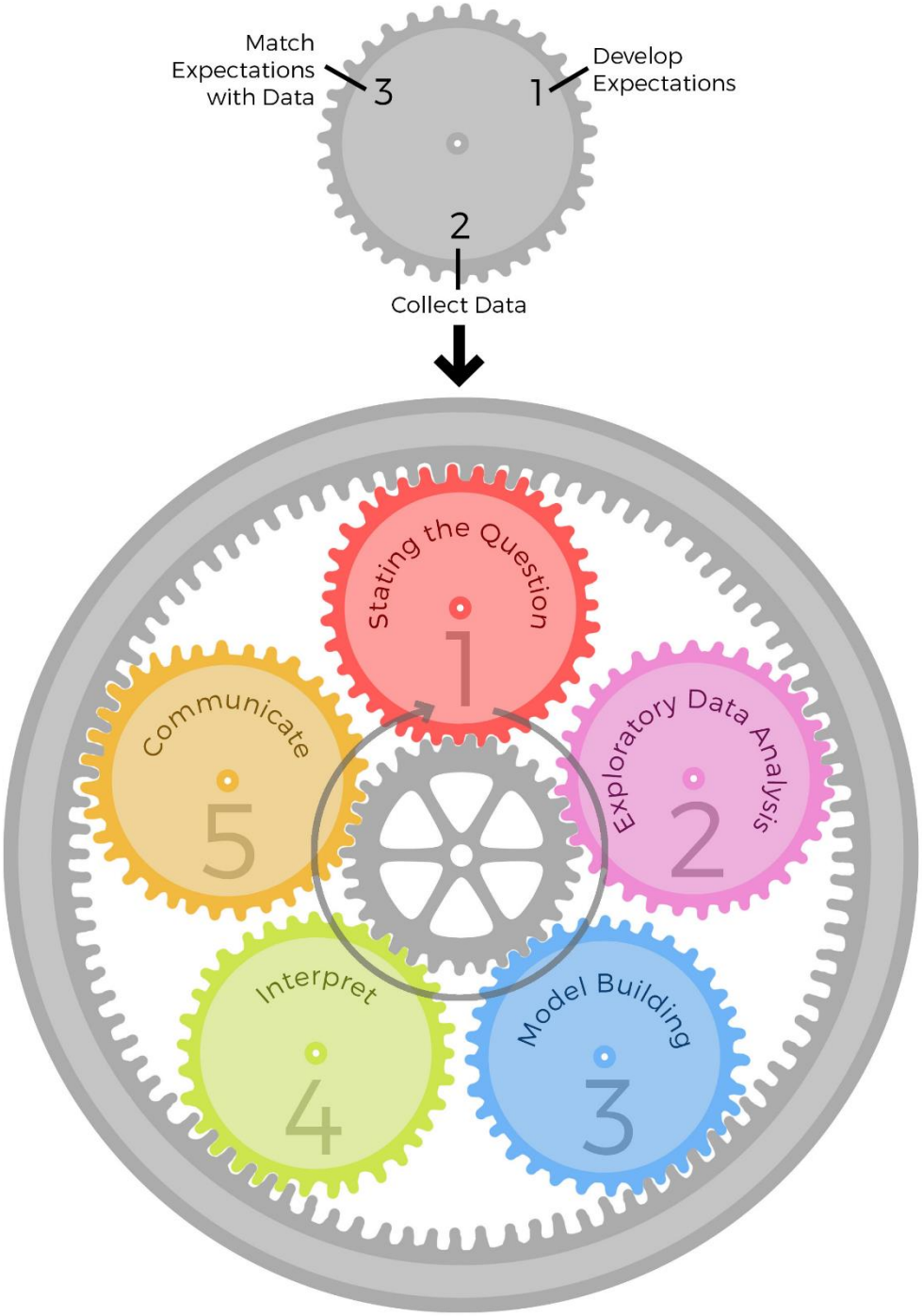
Data Model

The Art of Data Science

Epicycles of Analysis

- To the uninitiated, a data analysis may appear to follow a linear, one-step-after-the-other process .
- In reality, data analysis is a highly iterative and non-linear process, better reflected by a series of epicycles, in which information is learned at each step, which then informs whether (and how) to refine, and redo, the step that was just performed, or whether (and how) to proceed to the next step

https://bookdown.org/rdpeng/artofdatascience/epicycles-of-analysis.html



Statistical Model

A statistical model serves two key purposes in a data analysis,

- quantitative summary of data.
- impose a specific structure on the population from which the data were sampled.

A statistic is any summary of the data.

The sample mean, median, the standard deviation, the maximum, the minimum, and the range are statistics.

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Models as Expectations

- A statistical model must impose some structure on the data.
- **A statistical model provides a description of how the world works and how the data were generated.**
- The model is essentially an expectation of the relationships between various factors in the real world and in your dataset.
- Mimics the Population behavior, realized through Sample of data.
- A statistical model allows for some randomness in generating the data.

Data Model - Case Study

Conduct a survey of 20 people to ask them how much they'd be willing to spend on a product you're developing.

The survey response

25, 20, 15, 5, 30, 7, 5, 10, 12, 40, 30, 30, 10, 25, 10, 20, 10, 10, 25, 5

What do the data say?

Note: The example is hypothetical, generally we select higher sample size for modelling.

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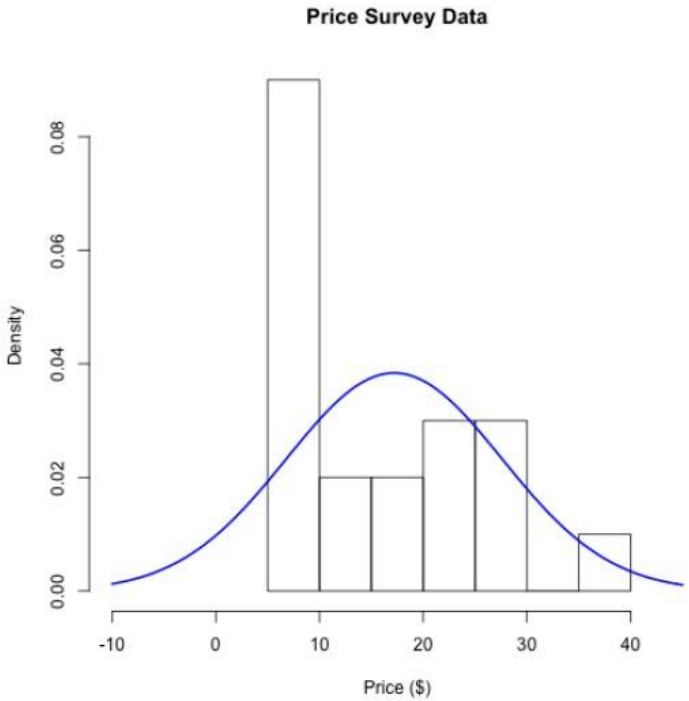
Step 1: Setting Expectations

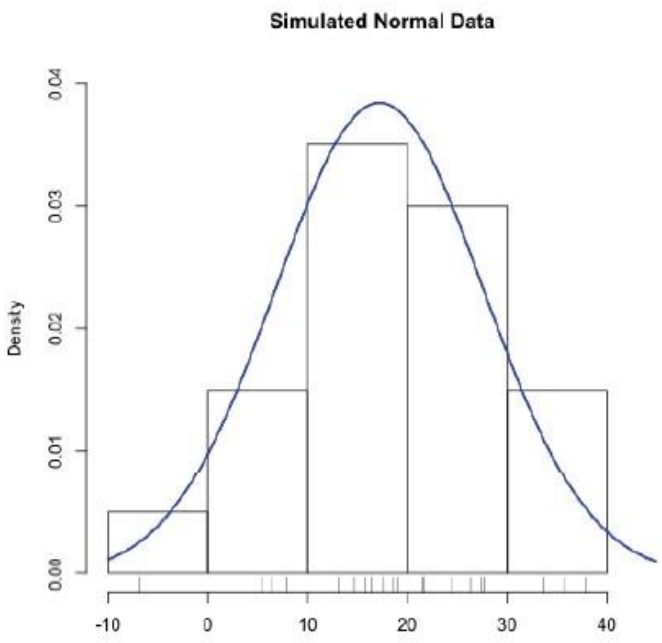
- The sample data represents the overall population likely to purchase the product.

  Mean - $17.2 and Standard Deviation - $10.39
- Expectation under the Normally distributed data (Normal model) is that the distribution of prices that people are willing to pay.
- According to the model, about 68% of the population would be willing to pay somewhere between $6.81 and $27.59 for this new product.



Step 2: Comparing Model Expectations with Reality

- Given the parameters, our expectation under the Normal model is that the distribution of prices that people are willing to pay looks like a bell-shaped curve.
- E.g. Normal curve on top of the histogram of the 20 data points of the amount people say they are willing to pay. The histogram has a large spike around 10.
- Normal distribution allows for negative values on the left-hand side of the plot, but there are no data points in that region of the plot.

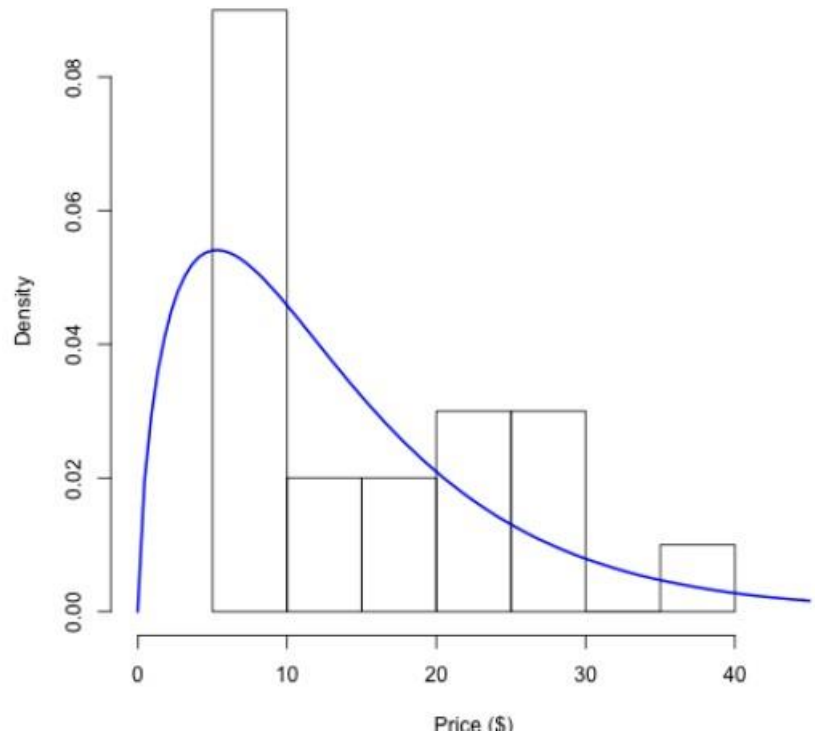


Histogram of Simulated Normal Data

Step 3: Refining Our Expectations

- When the model and the data don't match very well.
  - Get a different model.
  - Get different data.
  - Do both.
- E.g. Choose a different statistical model to represent the population, the Gamma distribution, which has the feature that it only allows positive.

$$\frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

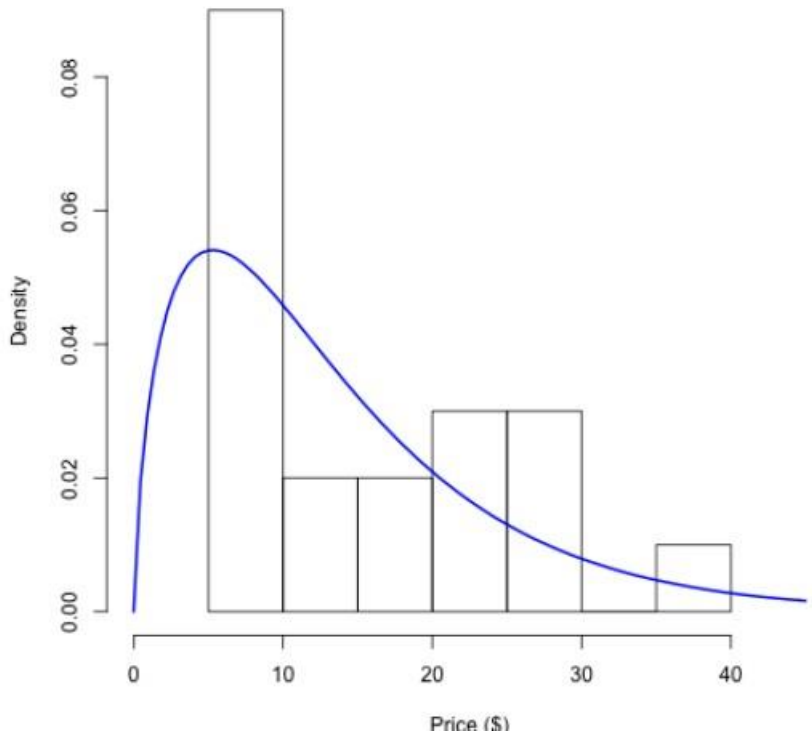


Mean = α/β

Variance = α/β^2

Step 3: Refining Our Expectations

- Normal vs Gamma Distribution – which model to choose?
- Qn 1: What percentage of the population is willing to pay atleast $30 for this product?
  - Normal distribution – 11% would pay $30 or more
  - Gamma distribution – can be less depending on choice of α and β
- Based on which model suits your problem at hand, choose the appropriate model.



Bonferroni's Principle

Total Information Awareness

- After the terrorist attack of Sept. 11, 2001, the US govt agencies were keen to use data to prevent.
- Four people enrolled in different flight schools, learning how to pilot commercial aircraft, although they were not affiliated with any airline.
- It was conjectured that the information needed to predict and foil the attack was available in data, but that there was then no way to examine the data and detect suspicious events.
- Total Information Awareness, a govt program, was intended to mine all the data it could find, including credit-card receipts, hotel records, travel data, and many other kinds of information in order to track terrorist activity.
- The program was later abandoned out of privacy concerns.

Bonferroni's Principle

- If you look for events of a certain type within data, the chance of finding them increases with the size of data, even if the data is completely random.

An Example of Bonferroni's Principle.

1. There are one billion people who are potential evil-doers.

2. Everyone goes to a hotel one day in 100.

3. A hotel holds 100 people. Hence, there are 100,000 hotels – enough to hold the 1% of a billion people who visit a hotel on any given day.

4. We shall examine hotel records for 1000 days.

- Hypothesis for detecting suspected evil-doers: Two people who, on two different days, were both at the same hotel.

Bonferroni's Principle

Probability of a person visiting a hotel = 0.01

Any two people both deciding to visit a hotel on any given day is $0.01*0.01 = 10^{-4}$

The probability of them visiting same hotel = $10^{-4} * 10^{-5} = 10^{-9}$

The probability that they(same two people) will visit the same hotel on two different given days = $10^{-9} * 10^{-9} = 10^{-18}$

Now, the number of pairs of people is ~ $5 * 10^{17}$

Now, the number of pairs of days is ~ $5 * 10^{5}$

The expected number of events that look like evil-doing is the product of the number of pairs of people, the number of pairs of days, and the probability that any one pair of people and pair of days is an instance of the behavior as per our hypothesis $= 5 * 10^{17} * 5 * 10^{5} * 10^{-18} = 250000$

Bonferroni's Principle

Let us change the hypothesis for detecting suspected evil-doers: Two people who, on three different days, were both at the same hotel.

The probability that they(same two people) will be in the same hotel on three different given days = $10^{-9} * 10^{-9} * 10^{-9} = 10^{-27}$

The expected number of events that look like evil-doing =

$5 * 10^{17} * 5 * 10^{5} * 10^{-27} = 2.5 * 10^{-4}$

Data Pipelines

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Data Pipeline Stages

Data pipelines are sets of processes that move and transform data from various sources to a destination where new value can be derived.

In their simplest form, pipelines may extract only data from one source such as a REST API and load to a destination such as a SQL table in a data warehouse.

In practice, data pipelines consist of multiple steps including data extraction, data preprocessing, data validation, and at times training or running a machine learning model before delivering data to its final destination.

Data engineers specialize in building and maintaining the data pipelines.

Why Build Data Pipelines?

- For every dashboard and insight that a data analyst generates and for each predictive model developed by a data scientist, there are data pipelines working behind the scenes.
- A single dashboard, or a single metric may be derived from data originating in multiple source systems.
- Data pipelines extract data from sources and load them into simple database tables or flat files for analysts to use. Raw data is refined along the way to clean, structure, normalize, combine, aggregate, and anonymize or secure it.

Data Pipeline Stages

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Data Ingestion

The term data ingestion refers to extracting data from one source and loading it into another.

Ingestion Interface and data structure

- A database behind an application, such as a Postgres or MySQL database or NoSQL database
- JSON from a REST API
- A stream processing platform such as Apache Kafka
- A shared network file system or cloud storage bucket containing logs, comma-separated value (CSV) files, and other flat files
- Semi-structured log data
- A data warehouse or data lake
- Data in HDFS or HBase database

Data ingestion is traditionally both the extract and load steps of an ETL or ELT

ETL and ELT

E– extract step

gathers data from various sources in preparation for loading and transforming.

L – load step

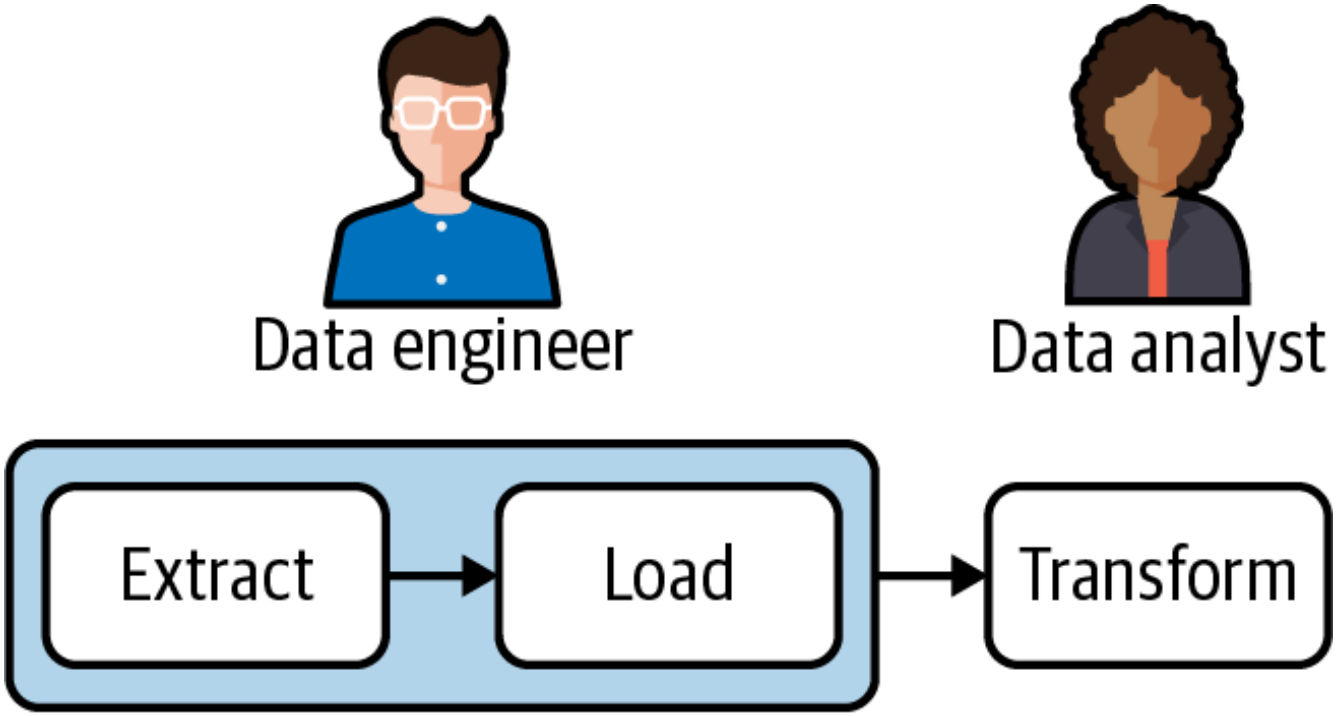
brings either the raw data (in the case of ELT) or the fully transformed data (in the case of ETL) into the final destination.

load data into the data warehouse, data lake, or other destination.

T – transform step

raw data from each source system is combined and formatted in a such a way that it's useful to analysts, visualization tools

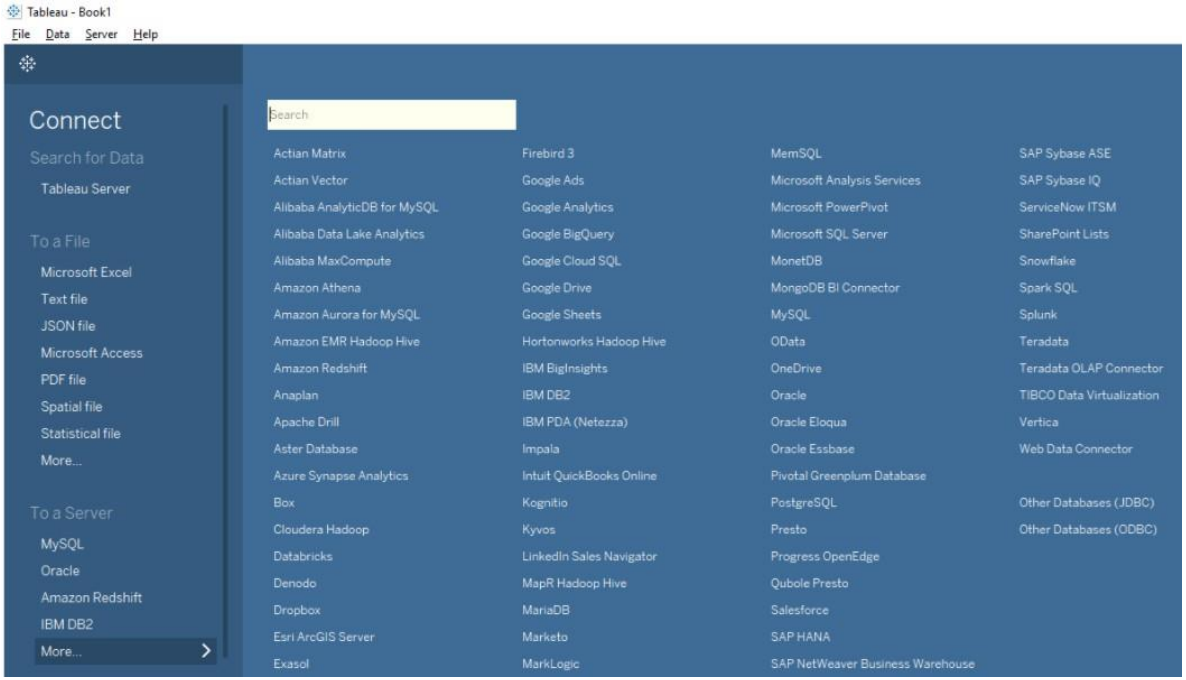
ELT Pipeline

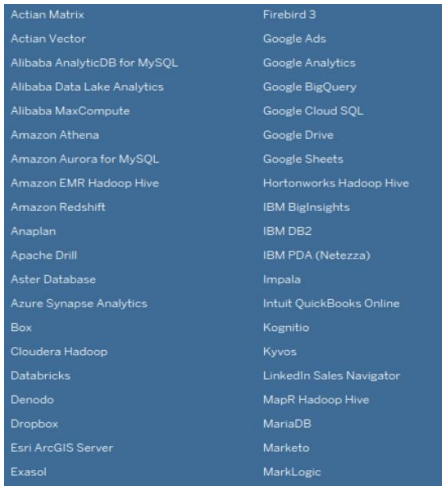


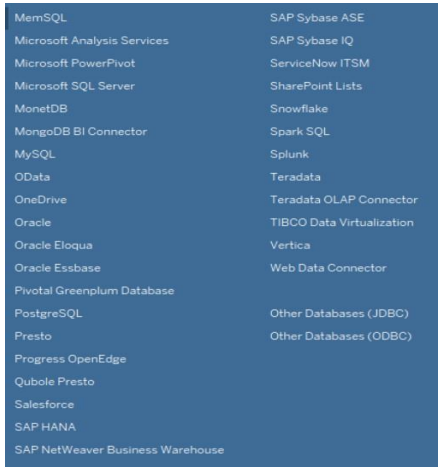
Orchestrating Pipelines

- Orchestration ensures that the steps in a pipeline are run in the correct order and that dependencies between steps are managed properly.
- Pipeline steps (tasks) are always directed, meaning they start with a task or multiple tasks and end with a specific task or tasks. This is required to guarantee a path of execution.
- Pipeline graphs must also be acyclic, meaning that a task cannot point back to a previously completed task.
- Pipelines are implemented as DAGs (Directed Acyclic Graphs). Orchestration tool – Apache Airflow

Various Data Sources







Text Books

T1	Introduction to Data Mining, by Tan, Steinbach and Vipin Kumar
T2	Introducing Data Science by Cielen, Meysman and Ali
T3	Storytelling with Data, A data visualization guide for business professionals, by Cole Nussbaumer Knaflic; Wiley
T4	Data Mining: Concepts and Techniques, Third Edition by Jiawei Han and Micheline Kamber Morgan Kaufmann Publishers

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Thank You